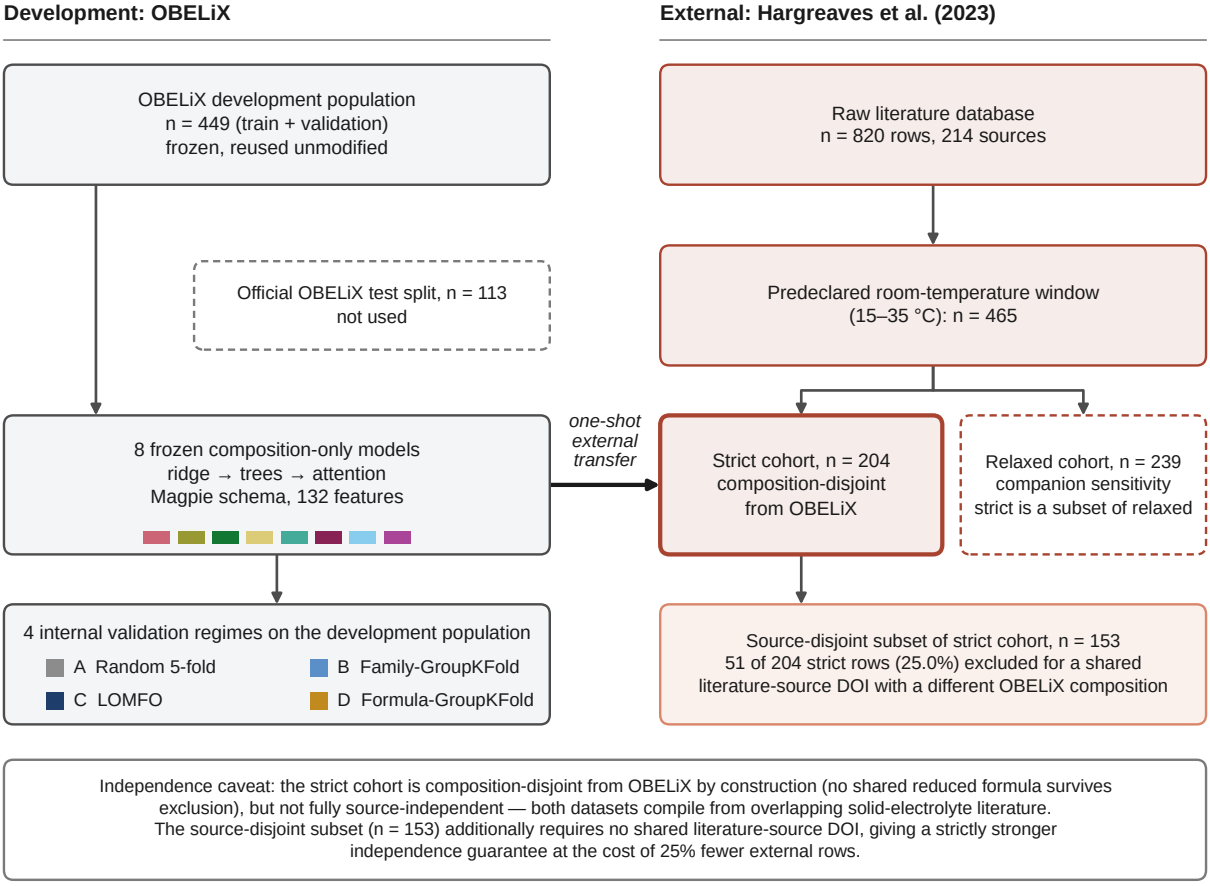
Development: OBELiX
External: Hargreaves et al. (2023)
OBELiX development population
n = 449 (train + validation)
frozen, reused unmodified
Official OBELiX test split, n = 113
not used
8 frozen composition-only models
ridge → trees → attention
Magpie schema, 132 features
one-shot
external
transfer
4 internal validation regimes on the development population
A Random 5-fold
B Family-GroupKFold
C LOMFO
D Formula-GroupKFold
Raw literature database
n = 820 rows, 214 sources
Predeclared room-temperature window
(15–35 °C): n = 465
Strict cohort, n = 204
composition-disjoint
from OBELiX
Relaxed cohort, n = 239
companion sensitivity
strict is a subset of relaxed
Source-disjoint subset of strict cohort, n = 153
51 of 204 strict rows (25.0%) excluded for a shared
literature-source DOI with a different OBELiX composition
Independence caveat: the strict cohort is composition-disjoint from OBELiX by construction (no shared reduced formula survives
exclusion), but not fully source-independent — both datasets compile from overlapping solid-electrolyte literature.
The source-disjoint subset (n = 153) additionally requires no shared literature-source DOI, giving a strictly stronger
independence guarantee at the cost of 25% fewer external rows.